

IA

INSTRUCTOR ASSISTANT

Interactive AI Assistant with
Voice and Text Integration

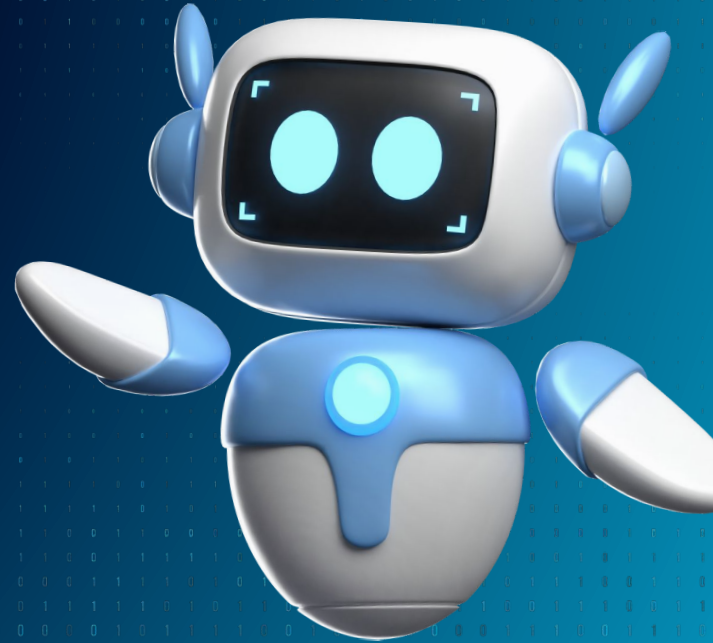
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Project Overview

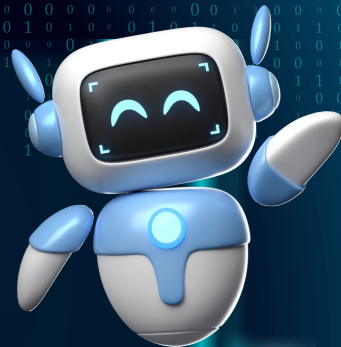
Create a dynamic chatbot using React (frontend) and Flask (backend).

- Features:

- Text-based chat and voice interaction.
- File upload and web link embedding.
- YouTube transcript processing.

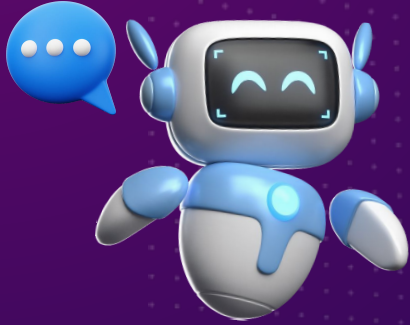
- Technology Stack:

- Frontend: React, Tailwind CSS, Firebase.
- Backend: Flask, OpenAI APIs, LangChain, Whisper, Google Cloud Speech API.
- Database: Firestore, Chroma.



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System Architecture



USERS INTERACT VIA
TEXT/VOICE.

FRONTEND (REACT) HANDLES
INPUT/OUTPUT.

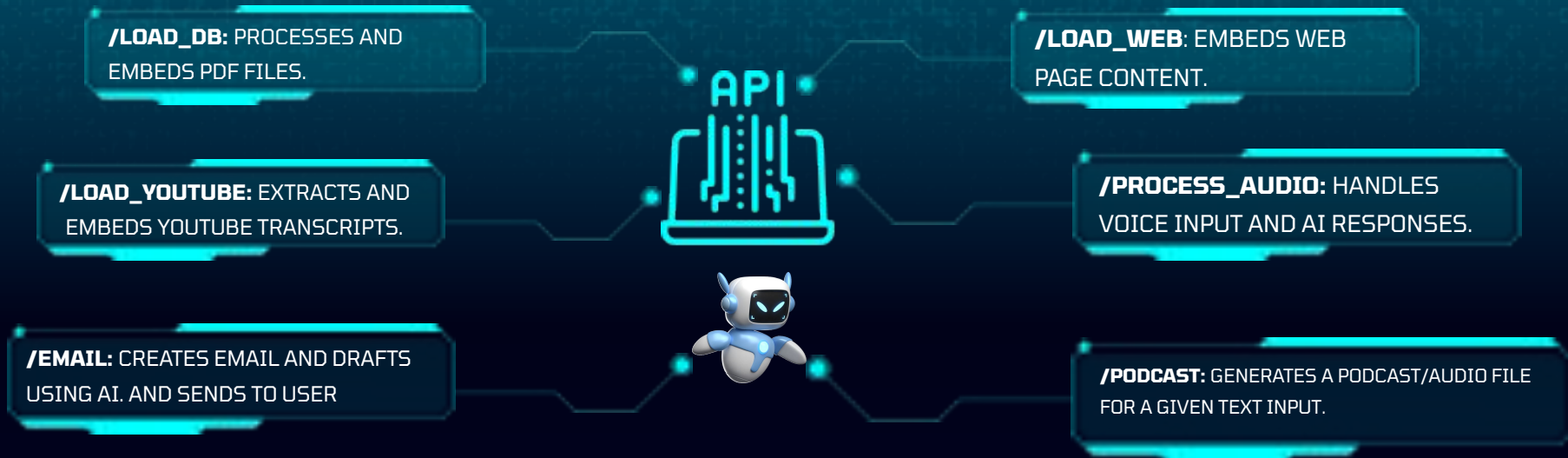
BACKEND (FLASK): PROCESSES REQUESTS
(FILES, URLS, AUDIO) AND COMMUNICATES
WITH OPENAI APIS.

RESPONSES DISPLAYED ON THE FRONTEND.

AI PROCESSING WITH LANGCHAIN AND
WHISPER.

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Backend Details



Libraries: Flask, LangChain, Whisper, Pydub.



Moderation System: Safeguarding Inputs

Ensure all user inputs are appropriate and safe for everyone.

Input Check:

- The system reviews what users type.
- Safe messages are processed normally.
- Unsafe messages are flagged with a friendly warning.

Flagged Message Response:

- Users see a helpful message asking them to revise their input.
- Problematic messages are saved securely for review.



AI Integration

OpenAI **Models**:

- GPT-3.5/4 for conversational responses.
- Whisper for voice transcription.
- Used about 987560 trained tokens for Fine Tuning

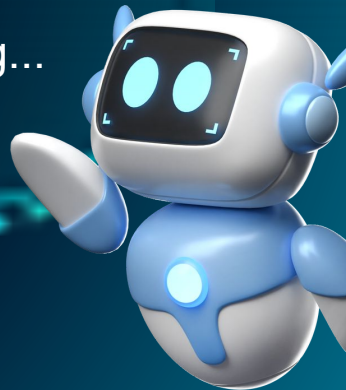
LangChain:

- Document processing, embedding, and retrieval.

Chroma Vector Store:

- Stores embeddings for efficient retrieval.

ai should
complement
teaching...



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Demo





Challenges:

- Real-time voice transcription.
- Managing large file uploads.
- Optimizing AI response times.

Solutions:

- Used Whisper for efficient transcription.
- Threading for non-blocking processes.
- LangChain for optimized embeddings.

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Future Enhancements



- Enhance AI models for more natural conversations.
- Support additional file formats (e.g., images, videos).
- Improve user authentication for private chats.
- Add Task specific AGENTS .



Conclusion

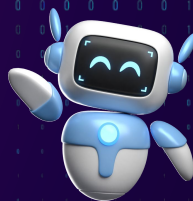
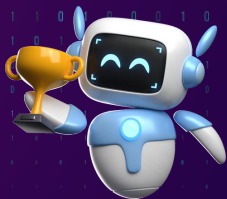
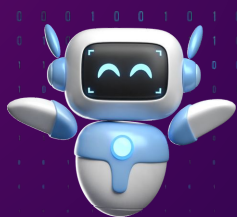
- Built a scalable chatbot platform integrating text and voice.
- Seamless interaction between frontend, backend, and AI.
- Demonstrated real-world use cases with advanced AI processing.

Google Slide :

https://docs.google.com/presentation/d/1457J8UuZK_4rIaUc9TDmMmy9AfC0qbl7UfGiwIxWpJU/edit?usp=sharing

Github:

<https://github.com/Montegan/IA.AI>



THANK YOU ANY QUESTIONS?

